# Corporate Finance

## Topic 2 – A

Bernardo Ricca

# Valuation

- We want to compute the price of an asset (project) $A$ today (at $t = 0$)
- The asset (project) generates a **random stream of future cash flows** $\tilde{d}_t$, where $t = 1, 2, 3, \ldots, T$

$$\begin{array}{l} \tilde{d}_1 \text{ at } t = 1 \\ \tilde{d}_2 \text{ at } t = 2 \\ \tilde{d}_3 \text{ at } t = 3 \\ \vdots \quad \vdots \quad \vdots \\ \tilde{d}_T \text{ at } t = T \end{array}$$

- In some cases, there is no clear final date $T$ (e.g. equity of a firm) and we let $T \to \infty$

# Valuation

- Examples: we can think of the asset (project) $A$ as
  - A new machine to improve the production process. In this case, $\tilde{d}_t$ is the **incremental free cash flow** generated by the new machine and $T$ is the length of its productive life
  - A new division of a firm (say a hotel chain expanding into the restaurant business)
- We want to find the value of the asset regardless of how it will be financed (the mix of debt and equity)

# Valuation

- The price of the asset, $P_A$, is the (expected) present value of the future cash flows

$$P_A = \sum_{t=1}^{T} \frac{\mathbb{E}[\tilde{d}_t]}{(1 + r_t)^t}$$

where

  - $\mathbb{E}[\tilde{d}_t]$ is the **expected cash flow** in period $t$
  - $r_t$ is the **expected rate of return** (discount rate)

- Note that the realized rate of return, $\tilde{r}_t$, is a random variable because the realized cash flows $\tilde{d}_t$'s are random

- Then, $\tilde{r}_t$ can be bigger or smaller than $r_t = \mathbb{E}[\tilde{r}_t]$

# Valuation

- Example: if $r_t$ is the $t$-year spot rate, the price of the following risk-free coupon bond (e.g., a Treasury Note)

| Year | 0 | 1 | 2 | ... | $N-1$ | $N$ |
|---|---|---|---|---|---|---|
| Cash flow | $-P$ | $C$ | $C$ | ... | $C$ | $C+F$ |

is given by

$$P = \frac{C}{1+r_1} + \frac{C}{(1+r_2)^2} + ... + \frac{C}{(1+r_{N-1})^{N-1}} + \frac{C+F}{(1+r_N)^N}$$

# Valuation

- Under the assumption of **time-invariant** discount rates, $r_t = r$ for all $t$, the formula simplifies to:

$$P_A = \sum_{t=1}^{T} \frac{\mathbb{E}[\tilde{d}_t]}{(1+r)^t}$$

# Valuation

- Challenges:
  - We need to measure and forecast expected future cash flows $\mathbb{E}[\tilde{d}_t]$
  - We need appropriate discount rates $r$
  - Appropriate discount rates come from an asset pricing model (e.g. CAPM, Fama-French) and consider the **riskiness** of the future cash flows $\tilde{d}_t$:

$$\text{discount rate} = \text{required expected return} =$$

$$\underbrace{\text{risk-free rate}}_{\substack{\text{compensation for} \\ \text{delaying consumption}}} + \underbrace{\text{risk-premium}}_{\substack{\text{compensation for} \\ \text{bearing risk}}}$$

# Discount rates

- CAPM: the expected discount rate or expected return is given by

$$r = \underbrace{\mathbb{E}[\tilde{r}]}_{\substack{\text{expected}\\ \text{return}}} = \underbrace{r_f}_{\substack{\text{risk-free}\\ \text{rate}}} + \underbrace{\underbrace{\beta}_{\substack{\text{systematic}\\ \text{risk}}} \underbrace{\left(\mathbb{E}[\tilde{r}_m] - r_f\right)}_{\substack{\text{market}\\ \text{risk premium}}}}_{\text{risk premium}}$$

where $\mathbb{E}[\tilde{r}_m]$ is the expected return on the market portfolio and[1]

$$\beta = \frac{Cov(\tilde{r}, \tilde{r}_m)}{Var(\tilde{r}_m)}$$

[1] Recall that the covariance of two random variables $X$ and $Y$ is

$$Cov(X, Y) = \mathbb{E}\Big[\big(X - \mathbb{E}[X]\big)\big(Y - \mathbb{E}[Y]\big)\Big] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

# Estimation of discount rates using the CAPM

- How do we estimate the discount rate using the CAPM?
  - Risk-free rate $r_f$: we usually use government bond yields at maturities representative of the duration of the project
  - Market risk premium $\mathbb{E}[\tilde{r}_m] - r_f$: we usually use slow-moving consensus approximations
  - Systematic risk $\beta$: for traded securities, we estimate it using past data
- But how do we estimate the $\beta$ of real investment projects for which we do not have historical data?

# Comparables

- One solution: find one (or more) publicly traded corporation with **similar assets** (business activities) and thus a similar market risk

- If the assets of this comparable firm have systematic risk $\beta_A$, then the systematic risk of the project is $\beta = \beta_A$ and the discount rate is

$$r = r_f + \underbrace{\beta}_{\substack{\text{systematic} \\ \text{risk of the} \\ \text{project}}} \left(\mathbb{E}[\tilde{r}_m] - r_f\right) = r_f + \underbrace{\beta_A}_{\substack{\text{systematic risk of} \\ \text{the comparable} \\ \text{firm's assets}}} \left(\mathbb{E}[\tilde{r}_m] - r_f\right)$$

# Comparables

- Problem: we can only estimate the systematic risk of the comparable firm's **equity**, $\beta_E$, as we only have historical stock prices
- If debt is publicly traded, we can also estimate the systematic risk of the comparable firm's **debt**, $\beta_D$
  - But this is not the case in general: bank debt, trade credit, illiquid bonds
  - One solution: look at bond indices with the same credit rating
  - When leverage is low and thus the default probability is very unlikely, we can assume that debt is risk-free and $\beta_D = 0$

# The relation between $\beta_E$ and leverage

- The risk of equity increases with leverage
- The equity risk $\beta_E$ of a highly levered firm overestimates the risk of its assets, $\beta_A$
- Simple example: house that is worth \$ 100 today and its price moves with the "market"
- Tomorrow, the market return can take values

$$\tilde{r}_m = \begin{cases} 10\%, & \text{with probability } 0.8 \\ -5\%, & \text{with probability } 0.2 \end{cases}$$

and therefore $r_m = \mathbb{E}[\tilde{r}_m] = 7\%$ and $Var[\tilde{r}_m] = 0.36\%$

# The relation between $\beta_E$ and leverage

- The risk-free rate is $r_f = 2\%$
- As the value of the house moves with the "market", $r_A = r_m = 7\%$, $\beta_A = 1$, and its value tomorrow, $\tilde{C}_A$, is

$$\tilde{C}_A = \begin{cases} 100 \times (1 + 10\%) = 110, & \text{with probability } 0.8 \\ 100 \times (1 - 5\%) = 95, & \text{with probability } 0.2 \end{cases}$$

# The relation between $\beta_E$ and leverage

- Case 1: house bought with 20% debt:

| | Value | Value tomorrow | | Return | | | |
|---|---|---|---|---|---|---|---|
| | today | Up | Down | Up | Down | Average | Beta |
| A | 100 | 110 | 95 | 10% | -5% | 7% | 1 |
| D | 20 | 20.4 | 20.4 | 2% | 2% | 2% | 0 |
| E | 80 | 89.6 | 74.6 | 12% | -7% | 8% | 1.25 |

- When $\frac{D}{D+E} = 20\%$, $r_E = 8\%$ and $\beta_E = 1.25$

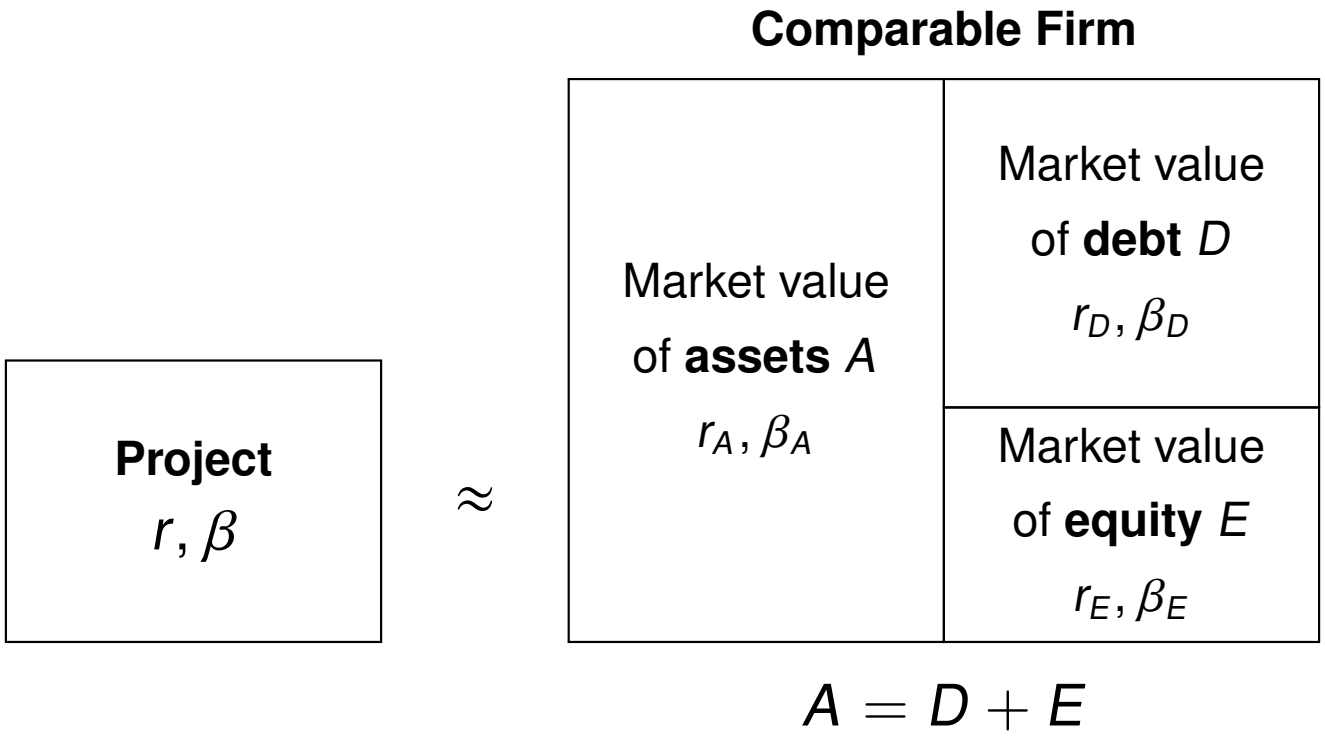
# The relation between $\beta_E$ and leverage

- Case 2: house bought with 50% debt:

| | Value | Value tomorrow | | Return | | | |
|---|---|---|---|---|---|---|---|
| | today | Up | Down | Up | Down | Average | Beta |
| A | 100 | 110 | 95 | 10% | -5% | 7% | 1 |
| D | 50 | 51 | 51 | 2% | 2% | 2% | 0 |
| E | 50 | 59 | 44 | 18% | -12% | 12% | 2 |

- When $\frac{D}{D+E} = 50\%$, $r_E = 12\%$ and $\beta_E = 2$
- In both cases, $r_E$ is greater than $r_A$

# Comparables




- $\beta_A$ is not observable but $\beta_E$ and $\beta_D$ are
- Can we estimate $\beta_A$ using $\beta_E$ and $\beta_D$?

# Unlevering $\beta_E$

- Yes: we need to "**unlever**" $\beta_E$
- Denote by $A$, $E$ and $D$ the market value of the assets, equity, and debt of the comparable firm
- As all the cash flows generated by assets are distributed to debholders and equityholders, investing \$1 in the assets of the firm is equivalent to investing in $\$D/(D+E)$ in the debt of firm and $\$E/(D+E)$ in the equity of the firm
- Therefore, the expected return on the assets is a weighted average of the expected return on the equity and debt

$$r_A = \frac{D}{D+E} r_D + \frac{E}{D+E} r_E$$

# Unlevering $\beta_E$

- An alternative way to derive the equation above is the following. Consider a one-period firm and denote by
  - $\tilde{C}_A$ the random cash flow generated by the firm's assets at $t = 1$
  - $\tilde{C}_D$ the repayment to debtholders
  - $\tilde{C}_E = \tilde{C}_A - \tilde{C}_D$ the shareholder's compensation (residual claimants)
- Then

$$D = \frac{\mathbb{E}[\tilde{C}_D]}{1 + r_D} \quad E = \frac{\mathbb{E}[\tilde{C}_E]}{1 + r_E} \quad A = \frac{\mathbb{E}[\tilde{C}_A]}{1 + r_A}$$

# Unlevering $\beta_E$

- As $A = E + D$ and $\tilde{C}_A = \tilde{C}_E + \tilde{C}_D$,

$$A = \frac{\mathbb{E}[\tilde{C}_A]}{1 + r_A} = \frac{\mathbb{E}[\tilde{C}_E] + \mathbb{E}[\tilde{C}_D]}{1 + r_A} = \frac{E(1 + r_E) + D(1 + r_D)}{1 + r_A}$$

$$\Rightarrow \; 1 + r_A = \frac{E(1 + r_E)}{A} + \frac{D(1 + r_D)}{A}$$

$$\Rightarrow \; r_A = \frac{E}{D + E} r_E + \frac{D}{D + E} r_D$$

# Unlevering $\beta_E$

- The equation above links **expected returns** of assets, debt, and equity, but it is also true for **realized returns**, since

$$D = \frac{\tilde{C}_D}{1+\tilde{r}_D} \quad E = \frac{\tilde{C}_E}{1+\tilde{r}_E} \quad A = \frac{\tilde{C}_A}{1+\tilde{r}_A}$$

- A similar derivation yields

$$\tilde{r}_A = \frac{E}{D+E}\tilde{r}_E + \frac{D}{D+E}\tilde{r}_D$$

# Unlevering $\beta_E$

- As covariance is a linear operator, and $A$, $D$, and $E$ are constants

$$\begin{aligned} Cov(\tilde{r}_A, \tilde{r}_m) &= Cov\left(\frac{E}{D+E}\tilde{r}_E + \frac{D}{D+E}\tilde{r}_D\ ,\ \tilde{r}_m\right) \\ &= \frac{E}{D+E}Cov(\tilde{r}_E, \tilde{r}_m) + \frac{D}{D+E}Cov(\tilde{r}_D, \tilde{r}_m) \end{aligned}$$

- Then

$$\begin{aligned} \beta_A &= \frac{Cov(\tilde{r}_A, \tilde{r}_m)}{Var(\tilde{r}_m)} = \frac{E}{D+E}\frac{Cov(\tilde{r}_E, \tilde{r}_m)}{Var(\tilde{r}_m)} + \frac{D}{D+E}\frac{Cov(\tilde{r}_D, \tilde{r}_m)}{Var(\tilde{r}_m)} \\ &= \frac{E}{D+E}\beta_E + \frac{D}{D+E}\beta_D \end{aligned}$$

- As with expected (and realized) returns, the systematic risk of assets is a weighted average of the systematic risk of debt and equity

# Example

- Suppose you identify three comparison publicly-traded firms
- For those firms, you observe the total book value of debt (proxy for its market value), $D$, and the market value of equity, $E$
- We assume the debt of the comparison firms is risk-free and the three firms provide an equally good comparison for the project we want to value
- The risk-free rate is $r_f = 4\%$ and the expected return on the market is $r_m = \mathbb{E}[\tilde{r}_m] = 12.4\%$
- From past returns, we estimate $\beta_E$

# Example

- As $\beta_D = 0, \beta_A = \frac{E}{D+E}\beta_E$:

| Comparable firm | $\beta_E$ | $D$ | $E$ | $E/(D+E)$ | $\beta_A$ |
|---|---|---|---|---|---|
| A | 0.75 | 0.004 | 0.096 | 0.96 | 0.72 |
| B | 1 | 2.3 | 7.7 | 0.77 | 0.77 |
| C | 1.08 | 0.21 | 0.79 | 0.79 | 0.85 |
| Average | | | | | 0.78 |

- The cost of the capital of the project is then

$$r = 4\% + \underbrace{0.78}_{\substack{\text{average beta of} \\ \text{the assets of the} \\ \text{comparable firms}}} (12.4\% - 4\%) = 10.56\%$$

# Which cash flows should we discount?

- As discussed in the first lecture, investors are interested in the amount of cash they can extract from the firm
- This amount of cash available to investors (debtholders and equityholders) is given by the **unlevered free cash flow** or **free cash flow to the firm** or simply **free cash flow**:

$$FCF_t = EBIT_t(1 - \tau) + Depreciation - CAPEX_t - \Delta NWC_t$$

  where $\tau$ is the effective tax rate

- Textbook definition of $NWC_t$ is Accounts Receivable + Inventories - Accounts Payable

# Which cash flows should we discount?

- In real-world applications, one might need to make small adjustments to the free cash flow formula to reflect the true amount of cash generated by the firm
- This computation of the free cash flow does not take into account subsidies that may be generated by financing choices (e.g. debt tax shield)
  - Those subsidies will be accounted for in the discount rates
- We will use projections of $FCF_t$ to estimate $\mathbb{E}(\tilde{d}_t)$

# Which cash flows should we discount?

- We estimate cash flows on an **incremental** basis
  - We want to measure the additional cash flow generated by the project
  - The additional cash flows are the difference between cash flows with versus without the project
- We have to take into account opportunity costs: the value of the resource in its best alternative use
- We should not consider sunk costs (unrecoverable costs for which the firm is already liable)
- We have to treat inflation consistently (nominal cash flows are discounted by nominal rates, while real cash flows are discounted by real rates)

# Which cash flows should we discount?

- When $T$ is large (long-lived assets), the analyst projects free cash flows for a handful of years and then computes a terminal value
  - Typically, after 5 or 10 years, we assume that either the company is liquidated (sold) or $FCF_t$ is growing at a constant rate in perpetuity
- Salvage value (SV): cash flow that the firm receives from liquidating its assets at some future date
  - SV = liquidation price – liquidation costs
  - The firm is taxed on SV – PPE (i.e. capital gain/loss on the book value of assets)
- We assume that NWC is recouped at the project end

# Project adoption rules: NPV

- We have built a framework to value (price) assets and projects
- When should we adopt a project?
- The net present value (NPV) project adoption rule states that *a project should be undertaken if its NPV is positive*
- For any asset (project) $A$ that requires an investment of $c_A$ at $t = 0$, the NPV is defined by

$$NPV_A = PV_A - c_A$$

# Project adoption rules: NPV

- $PV_A$: value of the stream of cash flows "inside the firm"
- $c_A$: cost of the stream of cash flows or value of the stream of cash flows "outside the firm"
- A positive NPV project represents an "arbitrage" across the boundaries of the firm
  - Unlike other hypothetical arbitrage opportunities, positive NPV projects are not scalable, so they can exist in equilibrium
- A positive NPV projects enrich (in expectation) the current claimants on the assets of the firm (equityholders and debtholders)
  - How equityholders and debtholders split the positive NPV can distort decisions (e.g. debt overhang, later in the course)

# Project adoption rules: NPV

- An example: Suppose firm $P$ (acquirer) is considering to buy out firm $T$ (target), which is currently held by a single shareholder $S$
  - If $S$ is willing to sell the target company $T$ for $c_T$, whereas, based on $P$'s valuation, the value of firm $T$ is $PV_T$, then the acquisition is a positive NPV investment if $PV_T > c_T$
  - Why does the valuation of the firm $T$ differ when it is operated as a separate entity owned by $S$, $c_T$, and when it forms a conglomerate with $P$?
  - It could be that $T$ being part of $P$ increases efficiency (synergies, economies of scale/scope, better management practices) or even market power gains

# NPV and shareholder wealth maximization

- In an **all-equity** firm, with **symmetric information**, the adoption of a positive NPV project increases the market value of equity by the NPV of the project
  - Not necessarily true in the presence of leverage (risky debt) and asymmetric information
- Let's assume that all agents are risk-neutral, in which case there is no risk premium and discount rates are equal to the risk-free rate for any project
- Alternatively, we can assume that the project we want to evaluate has no systematic risk ($\beta = 0$, all risks are idiosyncratic), again implying that the discount rate is equal to the risk-free rate

# NPV and shareholder wealth maximization

- Consider a company whose assets in place can take the following values at $t = 1$

| State of the economy | Probability | Market value of assets in place (t=1) |
|---|---|---|
| Boom | 0.5 | 100 |
| Slump | 0.5 | 10 |

- The company has access to a (safe) project that pays 22 in both states at $t = 1$ at a cost of 15 at $t = 0$
- Assume that the risk-free rate is 10% and that corporate taxes are zero
- The NPV of the project is -15+22/(1+10%)=5>0

# NPV and shareholder wealth maximization

- According to the NPV rule, the firm should go ahead with the project
- Suppose the current shareholders have the 15 required to fund the project and decide to adopt the NPV rule
- Without the project: the market value of equity is

$$\frac{0.5 \times 100 + 0.5 \times 10}{1 + 10\%} = 50$$

- With the project: the market value of equity is

$$\frac{0.5 \times (100 + 22) + 0.5 \times (10 + 22)}{1 + 10\%} = 70$$

# NPV and shareholder wealth maximization

- As shareholders had to pay 15, their gain is

$$\underbrace{(70 - 50)}_{\substack{\text{change in the} \\ \text{equity market value}}} - \underbrace{15}_{\text{outlay}} = 5 = NPV$$

- This conclusion does not rely on the project being funded by the shareholders

- If the firm decides to raise 15 by issuing additional equity in **competitive markets**, they have to sell a share $\alpha$ of the firm such that external investors make zero profits

$$\underbrace{\alpha \left( \frac{0.5 \times (100 + 22) + 0.5 \times (10 + 22)}{1 + 10\%} \right)}_{\text{value of their claim}} = \underbrace{15}_{\text{outlay}} \iff \alpha = \frac{15}{70}$$

# NPV and shareholder wealth maximization

- Initial shareholders retain a share $1-\alpha = 55/70$ of the firm with the project, which is worth

$$\frac{55}{70}\left(\frac{0.5\times(100+22)+0.5\times(10+22)}{1+10\%}\right)=55$$

- As shareholders did not have to pay 15, their gain is 55-50=5=NPV

- Similarly, if the firm decides to raise 15 by issuing (in this case, risk-free) debt in **competitive markets**, they promise to repay $D$ where

$$\frac{D}{1+10\%}=15 \iff D=16.5$$

# NPV and shareholder wealth maximization

- The shareholder's payoff is

$$\frac{0.5 \times (100 + 22 - 16.5) + 0.5 \times (10 + 22 - 16.5)}{1 + 10\%} = 55$$

- As shareholders did not have to pay 15, their gain is 55-50=5=NPV
- Conclusion: without frictions, that is, with symmetric information and competitive financial markets, and without leverage, the NPV adoption rule is equivalent to shareholder value maximization
  - The increase in shareholder value is equal to the NPV

# Other project adoption rules

- Two alternatives are payback and internal rate of return (*IRR*)
- But those two rules can lead to wrong decisions
- Consider the *IRR*, which is defined implicitly by

$$0 = -c_A + \sum_{t=1}^{T} \frac{\mathbb{E}[\tilde{d}_t]}{(1 + IRR)^t}$$

- *IRR* rule: undertake projects that have an IRR larger than the discount rate

# Other project adoption rules

- Pitfalls of *IRR*:
  - Multiple solutions
  - Non-existence of the *IRR*
  - It cannot distinguish lending vs borrowing

| Project | Cost ($t = 0$) | Cash flow ($t = 1$) | IRR | NPV (r=10%) |
|---|---|---|---|---|
| A | -1000 | 1500 | 50% | 363.6 |
| B | 1000 | -1500 | 50% | -363.6 |

  - Mutually exclusive projects (scale)

| Project | Cost (t=0) | Cash flow (t=1) | IRR | NPV (r=10%) |
|---|---|---|---|---|
| A | -10 | 14.3 | 43% | 3 |
| B | -1000 | 1210 | 21% | 100 |

# Financing choices

- Capital structure is the composition of the claims against a firm's assets
- The most common form of liability is debt
- Differences between equity and debt:
- **(i) Payoff structure**: debt contracts establish fixed promised payments at fixed dates; the payoff of equity contracts is variable in nature as equityholders are residual claimants

# Financing choices

- **(ii) Seniority**: Debt is senior to equity, that is, shareholders only get paid after fulfilling the promises to debtholders
- **(iii) Maturity**: Debt claims always have a fixed maturity, whereas equity claims are perpetual
- **(iii) Legal rights**: debtholders do not have control rights, that is, they do not have a say in the management of the firm; equity holders, on the other hand, have voting rights

# Financing choices

- Even though debtholders do not have control rights, they can impose restrictive **covenants** to protect their interests
- Examples: covenants restricting dividend payments, share repurchases, new debt issuance (especially secured or senior to current debt); covenants that transfer control in case conditions such as maximum leverage, minimum net worth (book value of assets minus book value of liabilities) or minimum liquidity, are violated
- Violations of covenants (other than the repayment of principal and interest) are referred to as **technical defaults**

# Financing choices

- If the borrower misses a payment (defaults), the lender can seize the firm's assets and assume the control of the firm, leaving equityholders empty-handed
- Under limited liability, only assets of the firm can be seized: owners' personal assets are protected
- In most jurisdictions, in case of default, debtholders can also report negative information about the borrower to a credit bureau

# Payoffs: equity vs debt

- Consider a firm (entrepreneur) that finances its operations at $t = 0$ with
  - Owner's own money (equity)
  - A one-period debt contract that specifies a fixed repayment of $F$ to creditors at maturity ($t = 1$)
- The value of the firm's assets is $\tilde{A}$ at $t = 1$
  - $\tilde{A}$ is unknown today: it might be bigger or smaller than $F$, that is, debt is risky

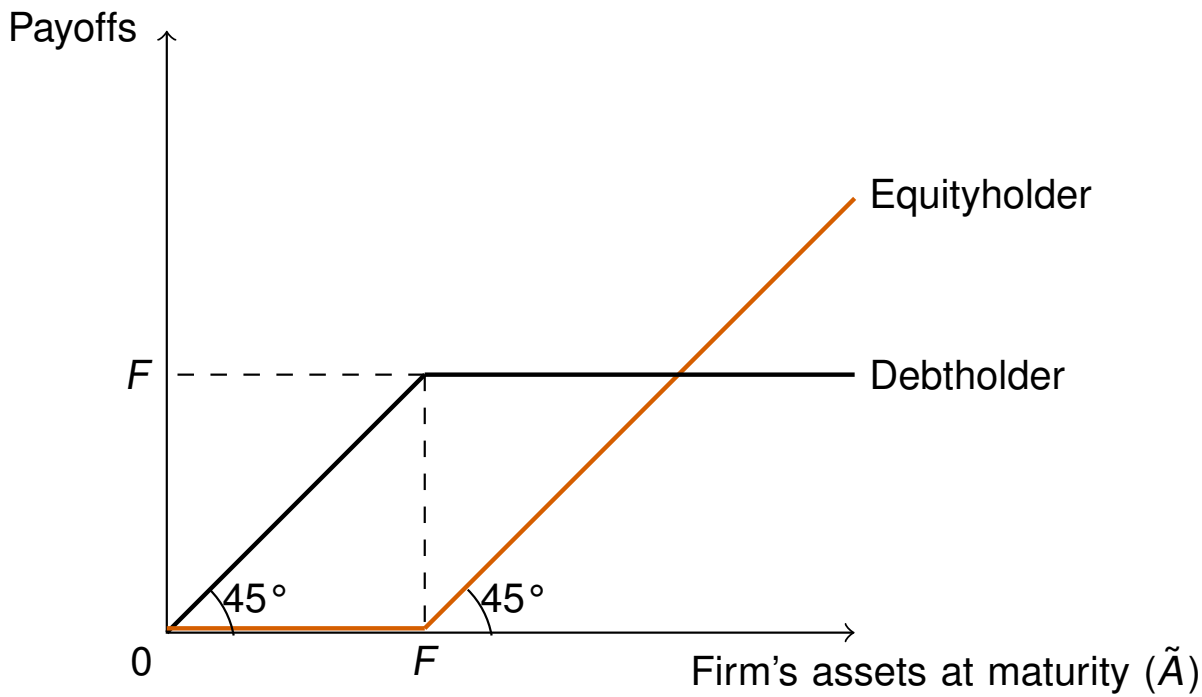
# Payoffs: equity vs debt

- We have two outcomes:
  - If $\tilde{A} < F$, the firm **defaults** on its debt. Creditors are senior and seize all the assets, receiving $\tilde{A}$; equityholders are empty-handed
  - If $\tilde{A} \geq F$, the firm can fulfill its promise to repay the fixed amount $F$; the equity value is the residual $\tilde{A} - F$
- At $t = 1$, the payoffs of debtholders and equityholders are (as a function of $\tilde{A}$)

| | Default $\tilde{A} < F$ | Full repayment $\tilde{A} \geq F$ |
|---|---|---|
| Debt | $\tilde{A}$ | $F$ |
| Equity | 0 | $\tilde{A} - F$ |

# Payoffs: equity vs debt

- Payoffs at maturity for a debt contract with no coupon payments and face value $F$




- Note that equity is a call option on the assets of the firm. We will see later that this feature can cause a conflict of interest between debtholders and equityholders

# A frictionless benchmark

- In a frictionless benchmark the capital structure is irrelevant
- The way the firm finances its assets does not affect their value
- This is known as the **Modigliani-Miller irrelevance result**
- In this world, there is no such a thing as an optimal capital structure (optimal mix of debt and equity)
- Strong assumptions (some of which we will relax later on), but a useful benchmark

# A frictionless benchmark: strong assumptions

- Competitive and complete markets
  - Individuals and firms are price-takers
  - Cash flow of (new) securities can be replicated by existing securities
- Individuals and firms can undertake financial decisions at the same price (e.g. borrow at the same rate)
- No taxes, bankruptcy, or other transaction costs
- Financing decisions neither affect cash flows generated by assets nor convey information
  - Fixes (exogenous) operating cash flow (fixed investments)
  - Symmetric information

# Digression: market completeness

- When the future is uncertain, we can think of possible scenarios as states of the world $s \in S = \{s_1, s_2, ..., s_N\}$
- Examples:
  - Tomorrow might be a rainy or a sunny day. Then $S = \{sun, rain\}$
  - GDP growth between today and one year from now might be -3%, 0%, or +3%. Then $S = \{boom, neutral, recession\}$
- A **contingent claim** is an asset that pays 1 in a given state $s \in S$ and zero otherwise

# Digression: market completeness

- In a one-period economy with future states $S = \{boom, neutral, recession\}$, we have three contingent claims:

| | Boom | Neutral | Recession |
|---|---|---|---|
| cc1 | 1 | 0 | 0 |
| cc2 | 0 | 1 | 0 |
| cc3 | 0 | 0 | 1 |

# Digression: market completeness

- We say the market is **complete** when it is possible to construct contingent claims for every possible state of the world
  - The contingent claim does not have to be traded explicitly
  - Investors only need to be able to synthesize or replicate all contingent claims by combining (buying and short-selling) traded securities
- **Law of One Price**: assets and combination of assets (portfolios) that have the same payoff should trade at the same price
  - Investors cannot make instantaneous profits by repackaging portfolios (there is no arbitrage)

# Digression: market completeness

- An arbitrage exists
  - If you can construct a portfolio that has a positive payoff today (inflow of cash) and does not require any payment back in the future, that is, a *free lunch*; or
  - If you can create a portfolio that has zero costs today but has a nonnegative payoff in every possible future scenario (strictly positive with positive probability), that is, a *money machine*
- An arbitrage does not exist in equilibrium, provided there is at least one rational investor who prefers more to less (non-satiation)
- In particular, assets with the same future stream of payoffs should have the same price (law of one price)

# Digression: market completeness

- Suppose the following two assets trade in a market with two states of the nature:

| | | Payoff ($t = 1$) | |
|---|---|---|---|
| | Price ($t = 0$) | Slump | Boom |
| Stock | 2 | 1.9 | 2.4 |
| Bond | 1 | 1.05 | 1.05 |

- The contingent claims are

| | | Payoff ($t = 1$) | |
|---|---|---|---|
| | Price ($t = 0$) | Slump | Boom |
| cc 1 | ? | 0 | 1 |
| cc 2 | ? | 1 | 0 |

# Digression: market completeness

- Is the market complete? Yes
  - We can replicate cc 1 by buying 2 units of the stock and short-selling 3.8/1.05 units of the bond
  - We can replicate cc 2 by short-selling 2 units of the stock and buying 4.8/1.05 units of the bond
- By the Law of One Price, the prices of the contingent claims are:

| | | Payoff ($t = 1$) | |
|---|---|---|---|
| | Price ($t = 0$) | Slump | Boom |
| cc 1 | 2*2-3.8/1.05 = 0.381 | 0 | 1 |
| cc 2 | 4.8/1.05-2*2 = 0.571 | 1 | 0 |

# Digression: market completeness

- When markets are complete we can find the price of any potential asset
- Consider the asset:

| | | Payoff ($t = 1$) | |
|---|---|---|---|
| | Price ($t = 0$) | Slump | Boom |
| Asset | ? | x | y |

- Buying this asset is equivalent to buying $y$ units of contingent claim 1 and $x$ units of contingent claim 2
- By the Law of One Price, the price of this asset is $0.381y + 0.571x$

# A frictionless benchmark: an example

- Consider a penniless entrepreneur that needs to raise 1 today ($t = 0$) from outside investors to set up a firm and invest in a project
- The project's payoff tomorrow ($t = 1$) depends on the aggregate state of the economy
  - Two equally probable states (50/50): slump and boom

| | | Payoff ($t = 1$) | | |
|---|---|---|---|---|
| | Investment (t=0) | Slump | Boom | Expected return |
| Project | 1 | 1.05 | 3.15 | 110% |

- Question: is $r_A = 110\%$? The investment of 1 is the book or the market value of assets?

# A frictionless benchmark: an example

- Two assets are already traded in this economy: a stock of another company and a government bond

| | | Payoff ($t = 1$) | | |
|---|---|---|---|---|
| | Price ($t = 0$) | Slump | Boom | Expected return |
| Stock | 1 | 0 | 2.8 | 40% |
| Bond | 1 | 1.05 | 1.05 | 5% |

- Investors can freely buy and sell (short) the stock and the bond

# A frictionless benchmark: an example

- Suppose the entrepreneur chooses to raise 1 by issuing debt
  - The entrepreneur can issue risk-free debt (zero default probability), as he/she can meet the payment of 1.05 (=1+5%) in both states
- Alternatively, the entrepreneur can issue equity and sell a share $\alpha$ of the firm to external investors in exchange for receiving 1 from new shareholders
- Any combination of debt and equity is also a possibility: raising $D$ by issuing debt and $1 - D$ by issuing equity
- Is there an optimal choice (that maximizes the firm value)?

# A frictionless benchmark: an example

- Payoffs at $t = 1$ under debt financing (borrowing 1 from creditors)

| | Slump | Boom |
|---|---|---|
| Total payoff | 1.05 | 3.15 |
| Debt | 1.05 | 1.05 |
| Entrepreneur | 0 | 2.1 |

- As debt is senior to equity, note that equityholders have a payoff of zero in the slump state

# A frictionless benchmark: an example

- Payoffs at $t = 1$ under equity financing

| | Slump | Boom |
|---|---|---|
| Total payoff | 1.05 | 3.15 |
| New shareholders | $\alpha 1.05$ | $\alpha 3.15$ |
| Entrepreneur | $(1-\alpha)1.05$ | $(1-\alpha)3.15$ |

  where $\alpha$ is share of the firm sold in exchange for receiving 1

- How can we compute $\alpha$?
  - Financial markets are competitive: the activity of providing financing should have a zero NPV (zero profits)
- How can we compute the market value of the assets and the entrepreneur's payoff?

# A frictionless benchmark: an example

- We can use the stock and the bond to find the no-arbitrage price of contingent claims:
  - Contingent claim 1 (cc 1): an asset that pays 1 in the boom state and zero otherwise; its price is denoted by $p_1$
  - Contingent claim 2 (cc 2): an asset that pays 1 in the slump state and zero otherwise; its price is denoted by $p_2$

| | | Payoff ($t = 1$) | |
|---|---|---|---|
| | Price ($t = 0$) | Slump | Boom |
| cc 1 | p1 | 0 | 1 |
| cc 2 | p2 | 1 | 0 |

- Even if those assets are not traded, their hypothetical prices should be consistent with **no-arbitrage**

# A frictionless benchmark: an example

- Note that 2.1 units of contingent claim 1 have the same payoff as the entrepreneur's position under debt financing

| | Slump | Boom |
|---|---|---|
| 2.1 units of contingent claim 1 | 0 | 2.1 |
| Entrepreneur's payoff under debt financing | 0 | 2.1 |

- As assets with the same payoff should have the same price, the value of the entrepreneur's position under debt financing is $2.1p_1$

# A frictionless benchmark: an example

- Note that $(1-\alpha)3.15$ units of contingent claim 1 together with $(1-\alpha)1.05$ units of contingent claim 2 have the same payoff as the entrepreneur's position under equity financing

| | Slump | Boom |
|---|---|---|
| $(1-\alpha)3.15$ units of cc 1 | 0 | $(1-\alpha)3.15$ |
| $(1-\alpha)1.05$ units of cc 2 | $(1-\alpha)1.05$ | 0 |
| Total | $(1-\alpha)1.05$ | $(1-\alpha)3.15$ |
| Entrepreneur's payoff under equity financing | $(1-\alpha)1.05$ | $(1-\alpha)3.15$ |

- As assets with the same payoff should have the same price, the value of the entrepreneur's position under equity financing is $(1-\alpha)3.15p_1 + (1-\alpha)1.05p_2$

# A frictionless benchmark: an example

- What about $\alpha$?
- Note that $\alpha 3.15$ units of contingent claim 1 together with $\alpha 1.05$ units of contingent claim 2 have the same payoff as the new shareholders' position
- Under symmetric information and competitive markets, investors make zero profits, that is, they undertake zero NPV transactions. Therefore,

$$\underbrace{1}_{\substack{\text{outlay} \\ \text{at } t=0}} = \underbrace{\alpha 3.15 p_1 + \alpha 1.05 p_2}_{\substack{\text{value of the} \\ \text{position at } t=0}}$$

# A frictionless benchmark: an example

- Finding $p_1$. We need to trade stocks and bonds to construct a portfolio with the same payoff as contingent claim 1
- If we buy $x$ stocks and $y$ bonds, the payoffs of this portfolio are

| | | Payoff ($t = 1$) | |
|---|---|---|---|
| | Quantity | Slump | Boom |
| Stock | $x$ | 0 | $2.8x$ |
| Bonds | $y$ | $1.05y$ | $1.05y$ |
| | Total | $1.05y$ | $2.8x + 1.05y$ |

- As the stock and the bond have a price of 1, the price of the portfolio is $x + y$. In general the price would be $xS + yB$, where $S$ is the stock price and $B$ is the bond price

# A frictionless benchmark: an example

- To replicate contingent claim 1, we must set $1.05y = 0$ and $2.8x + 1.05y = 1$.
- This yields $y = 0$ and $x = 1/2.8$
- Therefore, the price of contingent claim 1 is

$$p_1 = 0 \times 1 + \frac{1}{2.8} \times 1 = \frac{1}{2.8} = 0.357$$

# A frictionless benchmark: an example

- To replicate contingent claim 2, we must set $1.05y = 1$ and $2.8x + 1.05y = 0$ to replicate contingent claim 2
- This yields $y = 1/1.05$ and $x = -1/2.8$ (short position in the stock)
- Therefore, the price of contingent claim 2 is

$$p_2 = \frac{1}{1.05} \times 1 - \frac{1}{2.8} \times 1 = \frac{1}{2.8} = 0.595$$

# A frictionless benchmark: an example

- Remark: contingent claims 1 and 2 have the same expected payoff, yet they have different prices
- Reason: agents are risk-averse and the marginal utility of an additional dollar decreases as they get richer
- Therefore, agents prefer one dollar in the slump state (when they are poorer) than one dollar in the boom state (when they are richer)
- Contingent claim 1 (that pays 1 in the boom state) is cheaper than contingent claim 2 (that pays one in the slump state)

# A frictionless benchmark: an example

- Now that we have $p_1$ and $p_2$, we can calculate the market value of the firm and the entrepreneur's position under different financing choices
- The value of the entrepreneur's position under debt financing is $2.1p_1 = 0.75$ and the firm value (value of the assets) is $3.15p_1 + 1.05p_2 = 1.75$
- As lending is a zero NPV transaction due to competition, the market value of debt is equal to 1 (equal to the cash injection)
- Under debt financing, $A = 1.75$, $E = 0.75$, $D = 1$

# A frictionless benchmark: an example

- Under equity financing, the value of $\alpha$ is

$$\alpha = \frac{1}{3.15p_1 + 1.05p_2} = \frac{4}{7}$$

- The value of the entrepreneur's position under equity financing is

$$\frac{3}{7}3.15p_1 + \frac{3}{7}1.05p_2 = 0.75$$

- The value of the new shareholders' position is 1 (equal to their cash injection, zero NPV investment), so the total equity value is $0.75 + 1 = 1.75$

- Under equity financing, $A = E = 1.75$

# A frictionless benchmark: an example

- The firm value, $A = 1.75$, is independent of the financing choice
- This is an example of the Modigliani-Miller irrelevance result
- Competitive investors undertake zero NPV transactions
- The entrepreneur retains all the NPV of the project (0.75) regardless of the financing choice

# A frictionless benchmark: an example

- Under debt financing,

| | | Payoff ($t = 1$) | | |
|---|---|---|---|---|
| | Market value | Slump | Boom | Expected return |
| Debt (D) | 1 | 1.05 | 1.05 | 5% $(= r_D)$ |
| Equity (E) | 0.75 | 0 | 2.1 | 40% $(= r_E)$ |
| Assets (A) | 1.75 | 1.05 | 3.15 | 20% $(= r_A)$ |

- As expected, the relationship

$$r_A = \frac{E}{D+E} r_E + \frac{D}{D+E} r_D$$

is satisfied

# A frictionless benchmark: an example

- Under equity financing, $D = 0$, and

| | | Payoff ($t = 1$) | | |
|---|---|---|---|---|
| | Market value | Slump | Boom | Expected return |
| Equity (E) | 1.75 | 1.05 | 3.15 | 20% $(= r_E)$ |
| Assets (A) | 1.75 | 1.05 | 3.15 | 20% $(= r_A)$ |

- As expected, the relationship

$$r_A = \frac{E}{D+E} r_E + \frac{D}{D+E} r_D$$

is satisfied

# A frictionless benchmark: an example

- We can show that the financing choice is irrelevant for any amount borrowed $D \in [0, 1]$
- The financing choice being irrelevant means that the market value of the assets of the firm does not depend on $D$ and $E$
- As

$$A = \frac{\mathbb{E}[\tilde{C}_A]}{1 + r_A}$$

not only $A$, but also $r_A$ is independent of $D$ and $E$

- That is, for any financing choice, $r_A = 20\%$

# A frictionless benchmark: an example

- Recall that

$$r_A = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D$$

- Rearranging, we have that, for any choice of $D$ and $E$,

$$r_E = \frac{E+D}{E} \underbrace{r_A}_{\substack{\text{independent of} \\ D \text{ and } E \text{ and} \\ \text{equal to 20\%}}} - \frac{D}{E} r_D$$

- $r_E$ is then a function of $E$, $D$, and $r_D$

# A frictionless benchmark: an example

- In this example, for any financing choice, debt is risk-free (zero default probability), meaning that $r_D = 5\%$
- Therefore

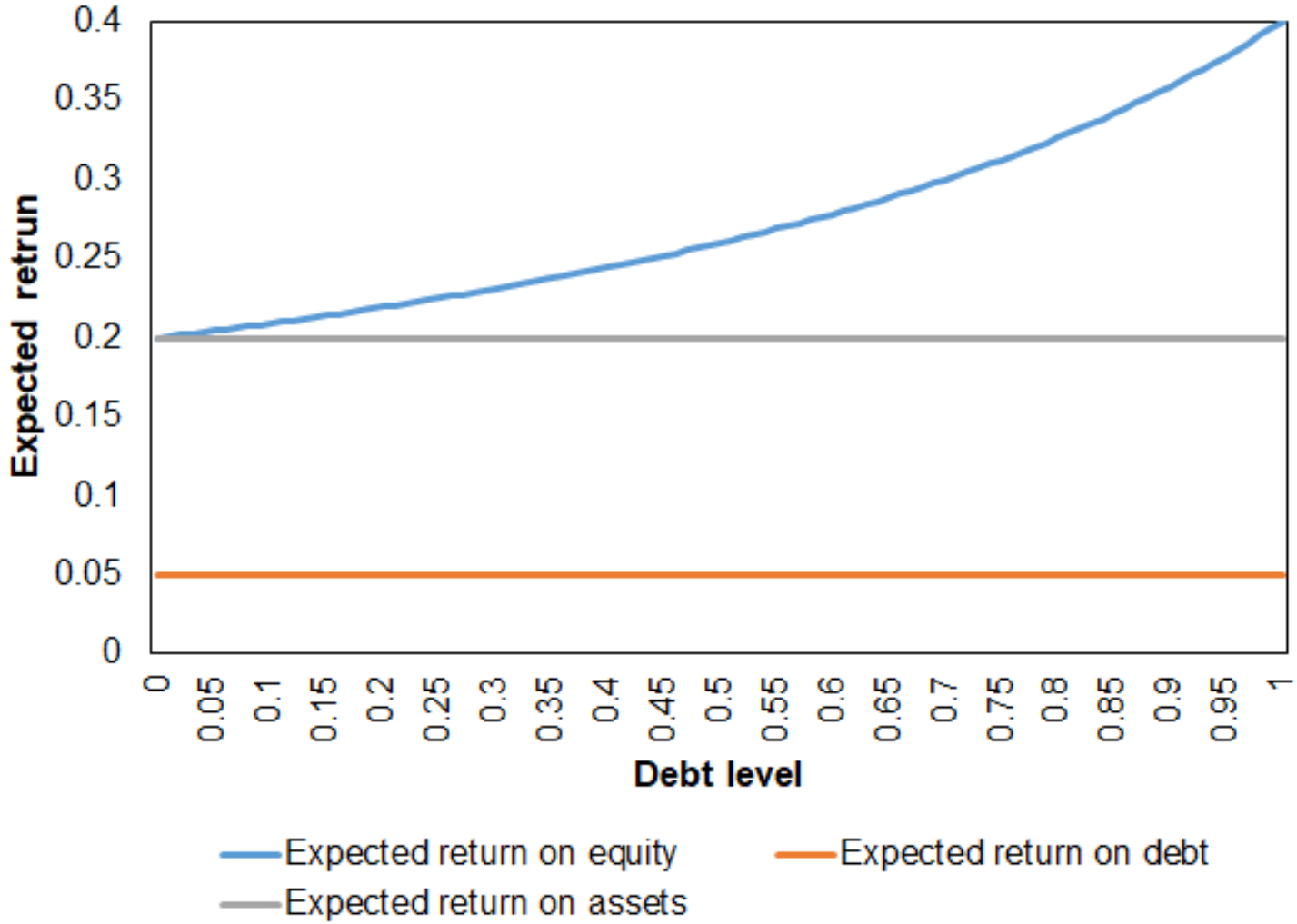
$$r_E = \frac{E+D}{E}20\% - \frac{D}{E}5\% = \frac{D}{E}15\% + 20\%$$

- As 1.75=E+D, for any $E$ and $D$,

$$r_E = \frac{D}{1.75 - D}15\% + 20\%$$

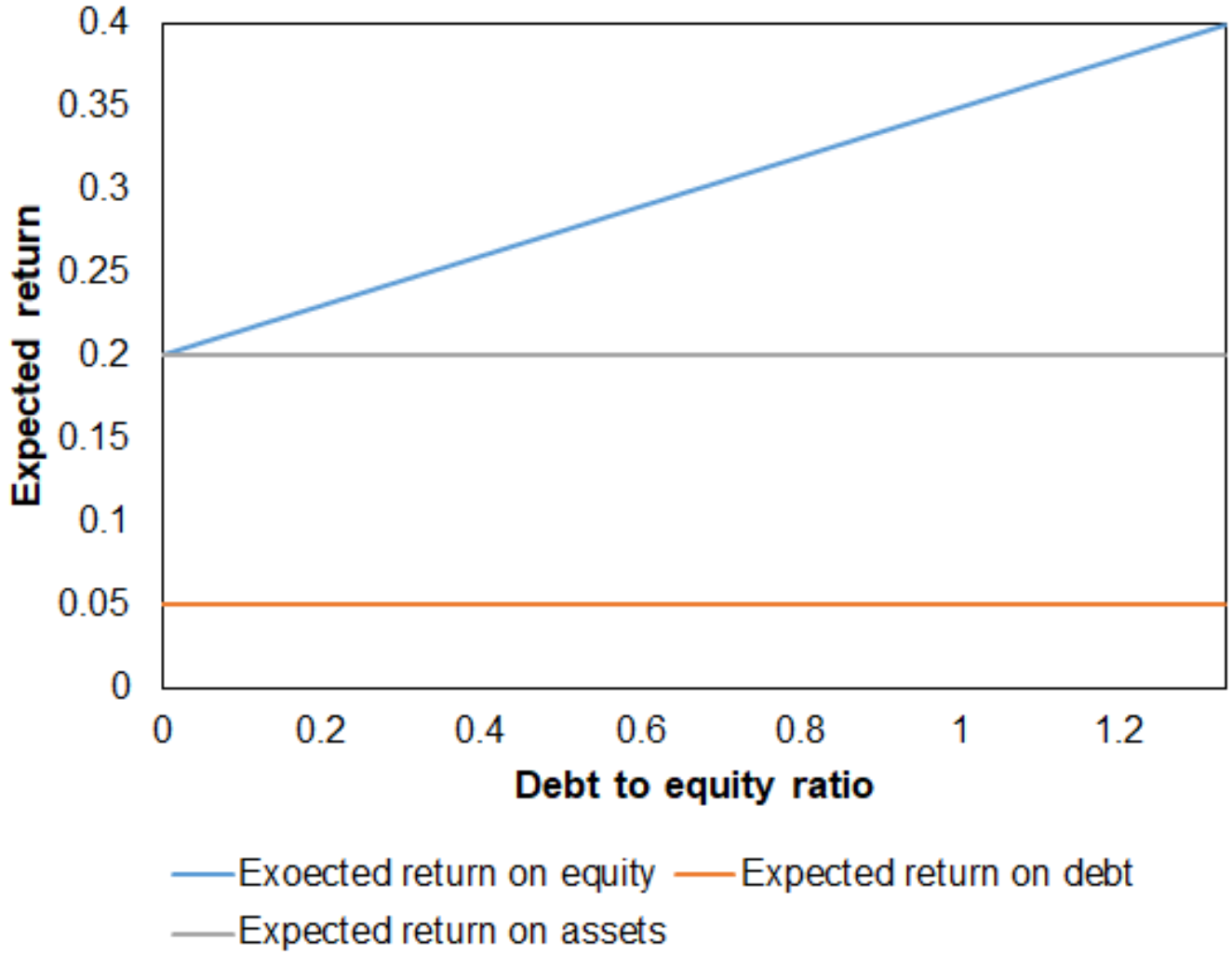
- $r_E$ increases with $D$: the more levered the firm, the higher the equity risk, and the higher the expected return $r_E$

# A frictionless benchmark: an example




- The expected return on equity, $r_E$, increases with the amount of debt, $D$

# A frictionless benchmark: an example




- The expected return on equity, $r_E$, increases with the debt to equity ratio, $D/E$. The increase is linear because debt is always risk-free in this example

# A frictionless benchmark: an example

- Suppose the entrepreneur decides to finance 0.4 out of the 1 with debt and the remainder 0.6 with equity
- Then

| | | Payoff ($t = 1$) | | |
|---|---|---|---|---|
| | Market value | Slump | Boom | Expected return |
| Debt | 0.4 | 0.42 | 0.42 | 5% $(= r_D)$ |
| Equity | 1.35 | 0.63 | 2.73 | 24.4% $(= r_E)$ |
| Assets | 1.75 | 1.05 | 3.15 | 20% $(= r_A)$ |

- As $r_D < r_E$, can the entrepreneur create value by borrowing more and using the proceeds to repurchase equity?

# A frictionless benchmark: an example

- Suppose the entrepreneur increases borrowing to $D = 0.5$
- If the expected returns do not change, we will obtain

| | | Payoff ($t = 1$) | | |
|---|---|---|---|---|
| | Market value | Slump | Boom | Expected return |
| Debt | 0.50 | 0.525 | 0.525 | 5% |
| Equity | 1.27 | 0.525 | 2.625 | 24.4% |
| Assets | 1.77 | 1.05 | 3.15 | |

- The value of the firm would increase
- Is this calculation correct? No! The expected return on equity must increase with leverage

# A frictionless benchmark: an example

- The correct market values are

| | | Payoff ($t = 1$) | | |
|---|---|---|---|---|
| | Market value | Slump | Boom | Expected return |
| Debt | 0.5 | 0.525 | 0.525 | 5% $(= r_D)$ |
| Equity | 1.25 | 0.525 | 2.625 | 26% $(= r_E)$ |
| Assets | 1.75 | 1.05 | 3.15 | 20% $(= r_A)$ |

- The expected return on equity increases from 24.4% to 26% when the amount borrowed increases from 0.4 to 0.5

- Again, this is an illustration of the Modigliani-Miller irrelevance result

# A frictionless benchmark: an example

- Exercise: we created contingent claims by trading the stock and the bond, and then we computed the value of the entrepreneur's position under debt and equity financing
- We could have replicated the entrepreneur's position under debt and equity financing directly by trading the stock and the bond
- You should arrive at the same results

# Modigliani-Miller: assumptions

- Competitive and complete markets
  - Individuals and firms are price-takers
  - Cash flow of (new) securities can be replicated by existing securities
- Individuals and firms can undertake financial decisions at the same price (e.g. borrow at the same rate)
- No taxes, bankruptcy, or other transaction costs
- Financing decisions neither affect cash flows generated by assets nor convey information
  - Fixes (exogenous) operating cash flow (fixed investments)
  - Symmetric information

# Modigliani-Miller proposition I (MM I)

- **(MM I)** *In a frictionless economy, total firm value is independent of the capital structure choice*
- Proof of the proposition (arbitrage argument)
  - Arbitrages cannot exist in equilibrium
  - Proof by contradiction: if two firms differing only in their capital structure have different market values, there exists an arbitrage opportunity

# MM I - proof sketch

- Consider two firms, firm 1 and firm 2
- The two firms are identical, except for their capital structure (the mix of debt and equity)
- At $t = 1$ both firms generate a random liquidating payoff $X$
  - We can think of $X$ as the resale value
- At $t = 0$, the firms have different capital structures:
  - Firm 1 has equity with a market value of $E_1$ and safe debt (zero default probability) with a market value of $D_1$
  - Firm 2 has no debt, $D_2 = 0$, and the market value of its equity is $E_2$

# MM I - proof sketch

- The market value of (the assets) of firm 1 is $V_1 = D_1 + E_1$
- The market value of (the assets) of firm 1 is $V_2 = E_2$
- Our goal is to show that $V_1 = V_2$
- The (gross) risk-free rate is $R_f = 1 + r_f$. At $t = 1$
- Complete markets imply that we can take a long (short) position on $E_1$, $D_1$, $V_1$, $E_2$, $V_2$ by combining securities that are currently traded and for which there is an equilibrium price

# MM I - case with safe debt

- At $t = 1$
  - Firm 1's debtholders receive $R_f D_1$
  - Firm 1's shareholders receive $X - R_f D_1$
  - Firm 2's shareholders receive $X$

# MM I - case with safe debt

- **Step 1**: Suppose that $V_2 > V_1$
- Arbitrage opportunity: same assets yet different market values
- An arbitrageur would sell the assets that are relatively overpriced and buy the ones that are relatively underpriced

| | Investment ($t = 0$) | Payoff ($t = 1$) |
|---|---|---|
| Short-sell a fraction $\alpha$ of firm 2's equity | $-\alpha V_2$ | $-\alpha X$ |
| Buy a fraction $\alpha V_2/V_1$ of firm 1's debt | $(\alpha V_2/V_1)D_1$ | $(\alpha V_2/V_1)R_f D_1$ |
| Buy a fraction $\alpha V_2/V_1$ of firm 1's equity | $(\alpha V_2/V_1)E_1$ | $(\alpha V_2/V_1)(X - R_f D_1)$ |
| Total | $\alpha[(V_2/V_1)(E_1 + D_1) - V_2]$ | $\alpha X(V_2/V_1 - 1)$ |
| | $= \mathbf{0}$ | $> \mathbf{0}$ |

# MM I - case with safe debt

- The arbitrageur can “undo firm 1’s leverage” by buying its debt and equity in the right proportion
- Remark:
  - A negative investment (negative price) means that there is an inflow of cash at $t = 0$
  - The investor has an inflow of cash at $t = 0$ because there is a commitment to repaying something (negative payoff) at $t = 1$
  - A positive investment (positive price) means that there is an outflow of cash at $t = 0$
  - The investor pays an amount at $t = 0$ because the other party in the transaction promises to repay something at $t = 1$ (the investor has a positive payoff)

# MM I - case with safe debt

- **Step 2**: Suppose that $V_1 > V_2$
- Again, an arbitrage opportunity:

| | Investment ($t = 0$) | Payoff ($t = 1$) |
|---|---|---|
| Buy a fraction $\alpha V_1 / V_2$ of firm 2's equity | $(\alpha V_1 / V_2) V_2$ | $(\alpha V_1 / V_2) X$ |
| Borrow $\alpha D_1$ | $-\alpha D_1$ | $-\alpha R_f D_1$ |
| Short-sell a fraction $\alpha$ of firm 1's equity | $-\alpha E_1$ | $-\alpha(X - R_f D_1)$ |
| Total | $\alpha[V_1 - (E_1 + D_1)]$ | $\alpha X(V_1 / V_2 - 1)$ |
| | $= \mathbf{0}$ | $> \mathbf{0}$ |

- The arbitrageur can "lever up" firm 2 by borrowing on individual accounts (homemade leverage)
- Absence of arbitrage implies that $V_1 = V_2$

# MM I - case with risky debt

- We consider a firm with debt that has a face value of $K$
- Risky debt means that default occurs with positive probability, that is, with positive probability the asset's payoff $X$ will be smaller than $K$
  - If debt were safe or risk-free, $X$ would always be bigger than $K$
- Risky debt with face value $K$ has a payoff of

$$\min\{K, X\} = \begin{cases} K, & \text{if } X \geq K \text{ (repayment of the fixed amount } K) \\ X, & \text{if } X < K \text{ (default: creditors seize all the assets)} \end{cases}$$

# MM I - case with risky debt

- We can rewrite the payoff of risky debt as

$$\min\{K, X\} = \underbrace{K}_{\substack{\text{payoff of a risk-free} \\ \text{bond with face value } K}} - \underbrace{\max\{K - X, 0\}}_{\substack{\text{payoff of a put on the} \\ \text{assets with strike } K}}$$

where

$$\max\{K - X, 0\} = \begin{cases} K - X & , \text{ if } X < K \\ 0 & , \text{ if } X \geq K \end{cases}$$

- Risky debt with face value $K$ is equivalent to a long position in a risk-free bond with face value $K$ and a short position in a put option on the assets of the firm with strike $K$

# MM I - case with risky debt

- Denote the no-arbitrage price of a put option with strike $K$ by $P(K)$
- The market value of risky debt is

$$D = \underbrace{\frac{K}{1+r_f}}_{\substack{\text{price of a risk-free} \\ \text{bond with face value } K}} - \underbrace{P(K)}_{\substack{\text{price of a put on the} \\ \text{assets of the firm with strike } K}}$$

- For a risk-free bond, $P(K) = 0$, as all the possible realizations of $X$ are bigger or equal than $K$
- We can think of $P(K)$ as the price discount of risky debt in comparison to risk-free (zero default probability) debt with the same face value

# MM I - case with risky debt

- The equity payoff is

$$\max\{X-K,0\} = \begin{cases} X-K & , \text{ if } X \geq K \text{ (repayment of the fixed amount } K) \\ 0 & , \text{ if } X < K \text{ (default: creditors seize all the assets)} \end{cases}$$

- This is the same payoff of a long position in a call option on the assets of the firm, $X$, with strike $K$
- Denote the no-arbitrage price of a call option with strike $K$ by $C(K)$
- The market value of equity must satisfy $E = C(K)$

# MM I - case with risky debt

- The firm's market value (the value of its assets) is

$$V(K) = \underbrace{C(K)}_{=E} + \underbrace{\frac{K}{1+r_f} - P(K)}_{=D}$$

- This is the **put-call parity**
  - $V(K)$ does not depend on $K$
  - Equivalently, the value of the assets of the firm is independent of the capital structure (MM I)

# MM I - case with risky debt

- Pick any $K_1$ and $K_2$
- Consider the following portfolios

Table: Portfolio 1

| | | Payoff ($t = 1$) | |
|---|---|---|---|
| | Investment ($t = 0$) | $X < K_1$ | $X \geq K_1$ |
| Buy a call on $X$ with strike $K_1$ | $C(K_1)$ | 0 | $X - K_1$ |
| Sell (write) a put on $X$ with strike $K_1$ | $-P(K_1)$ | $-(K_1 - X)$ | 0 |
| Buy a bond with face value $K_1$ | $K_1/(1 + r_f)$ | $K_1$ | $K_1$ |
| Total | $V(K_1)$ | $X$ | $X$ |

# MM I - case with risky debt

Table: Portfolio 2

| | | Payoff ($t = 1$) | |
|---|---|---|---|
| | Investment ($t = 0$) | $X < K_2$ | $X \geq K_2$ |
| Buy a call on $X$ with strike $K_2$ | $C(K_2)$ | 0 | $X - K_2$ |
| Sell (write) a put on $X$ with strike $K_2$ | $-P(K_2)$ | $-(K_2 - X)$ | 0 |
| Buy a bond with face value $K_2$ | $K_2/(1 + r_f)$ | $K_2$ | $K_2$ |
| Total | $V(K_2)$ | $X$ | $X$ |

- As these two portfolios have the same payoff, by non-arbitrage, they must have the same value, $V(K_1) = V(K_2)$: the firm market value (the market value of its assets) does not depend on $K$

# MM II

- How can we reconcile the fact that the mix of debt and equity is irrelevant with the fact that debt demands a lower return than equity?
- As we saw in the example, issuing more debt has a cost: it makes equity riskier and, as a result, more costly
- Even risk-free debt makes equity riskier (the case of our example): it is not (only) about default risk
- The costs of debt and equity are intertwined
- **(MM II)** *A firm's cost of equity increases as the market value debt to equity ratio increases (for low levels of debt, debt is risk-free, and the increase is linear)*

# MM II

- MM I implies that the market value of the firm, $V$, or equivalently, the market value of the assets of the firm, $A$, is independent of the choices of $D$ and $E$
- As the expected return on assets is defined by

$$A = \frac{\mathbb{E}[\tilde{C}_A]}{1 + r_A}$$

and $A$ and $\tilde{C}_A$, the cash flows generated by the assets, are independent of $D$ and $E$, $\boldsymbol{r_A}$ **is independent of** $\boldsymbol{D}$ **and** $\boldsymbol{E}$

# MM II

- Recall the identity

$$r_A = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D$$

- Rearranging, we have that

$$r_E = r_A + \frac{D}{E}(r_A - r_D)$$

- As $r_A$ is fixed (independent of $E$ and $D$) and $r_A > r_D$, $r_E$ increases with $D/E$, the market debt to equity ratio

- For small values of $D$, that is, when debt is risk-free, $r_D$ is fixed and equal to $r_f$, and $r_E$ is linear in $D/E$

# MM II

- When debt becomes larger, debt might become risky (positive default probability)
- In this case, $r_D$ also increases in $D/E$
- Then

$$\frac{\partial r_E}{\partial (D/E)} = r_A - \frac{\partial r_D}{\partial (D/E)}$$

- With a finite number of states, it can be the case that these effects offset each other: $r_E$ is strictly increasing in $D/E$ and then it becomes flat while $r_D$ increases in $D/E$

# MM II

- We refer to $r_A$ as the weighted average cost of capital, $r_{WACC}$
- MM I implies that $r_{WACC}$ does not depend on how the firm finances its assets (on the choices of $D$ and $E$)
- Another way to state MM I is

$$D_L + E_L = \underbrace{V_L}_{\substack{\text{value of the levered firm} \\ \text{(arbitrary values for } D_L \text{ and } E_L\text{)}}} = \underbrace{V_U}_{\substack{\text{value of the unlevered,} \\ \text{all-equity firm } (D = 0)}} = E_U$$

# MM II

- By MM I,

$$r_A = r_{WACC} = \frac{D}{D+E} \underbrace{r_E}_{\substack{\text{expected return} \\ \text{on equity of the} \\ \text{levered firm}}} + \frac{E}{D+E} r_D = \underbrace{r_U}_{\substack{\text{expected return} \\ \text{on equity of the} \\ \text{unlevered firm } (D=0)}}$$

- Then

$$r_E = r_U + \frac{D}{E}(r_U - r_D)$$

- **Conservation of value principle**: *With perfect capital markets, financial transactions neither add nor destroy value, but instead represent a repackaging of risk (and therefore return).*

# Comment on $r_D$

- $r_D$ is not the interest rate (contracted in the past) the firm pays on its debts
- Why?
  - Changes in macroeconomic conditions
  - Changes in firm's fundamentals (e.g. default risk)
- $r_D$ is not the yield to maturity

# Comment on $r_D$

- Consider a firm that has debt in place that matures in one period with face value $F$
- The **yield to maturity** $y$ is the discount rate that makes the present value of **promised** payments equal to the debt's market price:

$$\underbrace{D}_{\substack{\text{market value} \\ \text{of debt}}} = \frac{F}{1+y}$$

# Comment on $r_D$

- The **expected return on debt** $r_D$ is the discount rate that makes the present value of **expected** payments equal to the debt's market price:

$$\underbrace{D}_{\substack{\text{market value} \\ \text{of debt}}} = \frac{F \times (1-p) + C \times p}{1 + r_D}$$

  where $p$ is the probability of default and $C < F$ is the expected recovery in the event of default

- Whenever debt is risky, $y > r_D$: the yield overestimates the debt cost of capital